The Van der Pol Oscillator: Relaxation Cycles, Verified, and the Cost of Stiffness

SIRPY Problem Library

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SIRPY · Problem 17

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

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The problem

$$y'' = \mu\,(1-y^2)\,y' - y, \qquad y(0) = 2, \quad y'(0) = 0$$

Van der Pol's equation is an oscillator whose damping changes sign: small swings are pumped up, large ones damped down, and every trajectory settles onto one closed orbit. It was the first self-sustained oscillation understood mathematically, and it is still the standard example of one.

The parameter μ decides how violent that orbit is — and how punishing the equation is to integrate. At $\mu = 10$ it is brisk. At $\mu = 1000$ it is one of the standard **stiff** benchmarks: a fast mode forces any step-based method into extremely small steps, and cost explodes while the answer barely changes.

We ran three values on the same interval, changing one number.

What we handed the solver

```
solve_ivp(order=2, f_str=f"{mu}*(1 - y**2)*y1 - y",
          x0=0, x1=10, ic={0: 2, 1: 0})
```

The equation as written. The initial conditions as written. No Jacobian, no stiffness flag, no tolerance, no step size, no initial guess.

$\mu = 10$

```
Piecewise power-series solution (46 segments of degree 20 covering [0, 10])
  segment 1: [0, 0.0570568]  centered at 0
  segment 2: [0.0570568, 0.130555]  centered at 0.0570568
  ...

INFO: verification by substitution - VERIFIED: satisfies the equation
      to 6.17e-08 on [0, 10] (the equation's scale reaches 156; max
      RELATIVE residual, taken pointwise, is 1.00e-08). Checked by
      exact coefficient differentiation at interior points, not at the
      interface nodes. Interface continuity across 45 node(s): max
      one-sided jump in (y..y1) = 4.1e-10 - measured, not assumed.
      Also checked: initial data to 0.00e+00.
solve time: 0.130 s
```

Forty-six exact Taylor polynomials, joined end to end, in an eighth of a second. Not samples on a grid — **analytic pieces** you can differentiate, evaluate anywhere, and substitute back into the equation.

Notice the parenthesis. The right-hand side of this equation reaches a magnitude of 156, so an absolute residual of 6×10^{-8} means little on its own. SIRPY says so and gives the relative figure, 1.0×10^{-8}, unprompted.

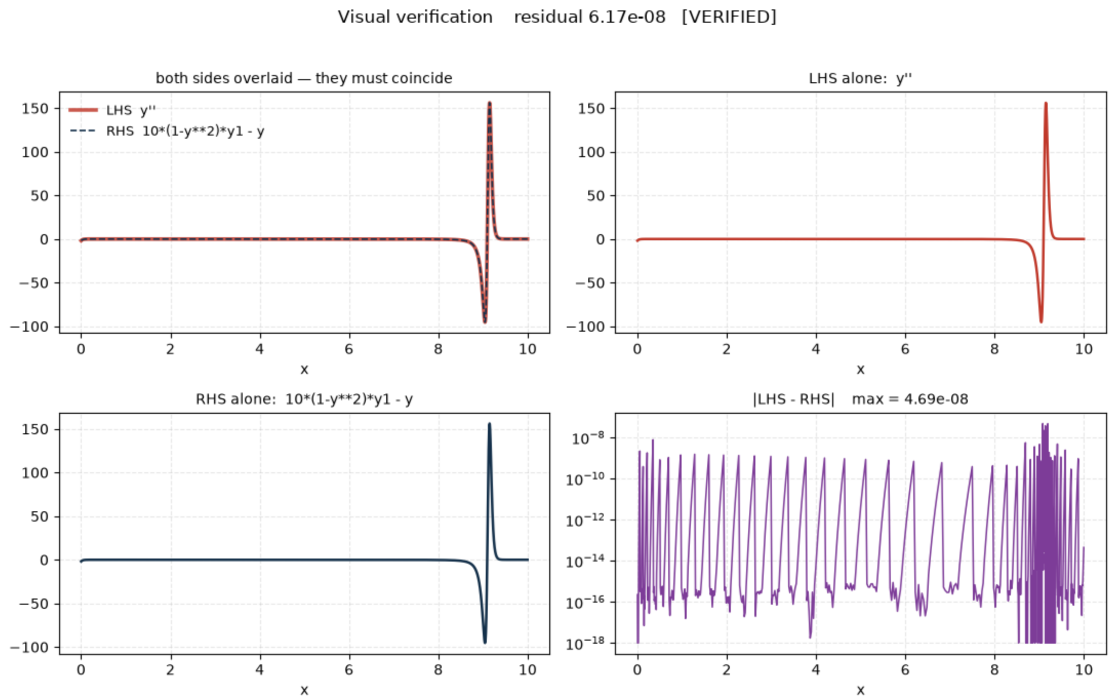

Figure 1: verification: $\mu = 10$.

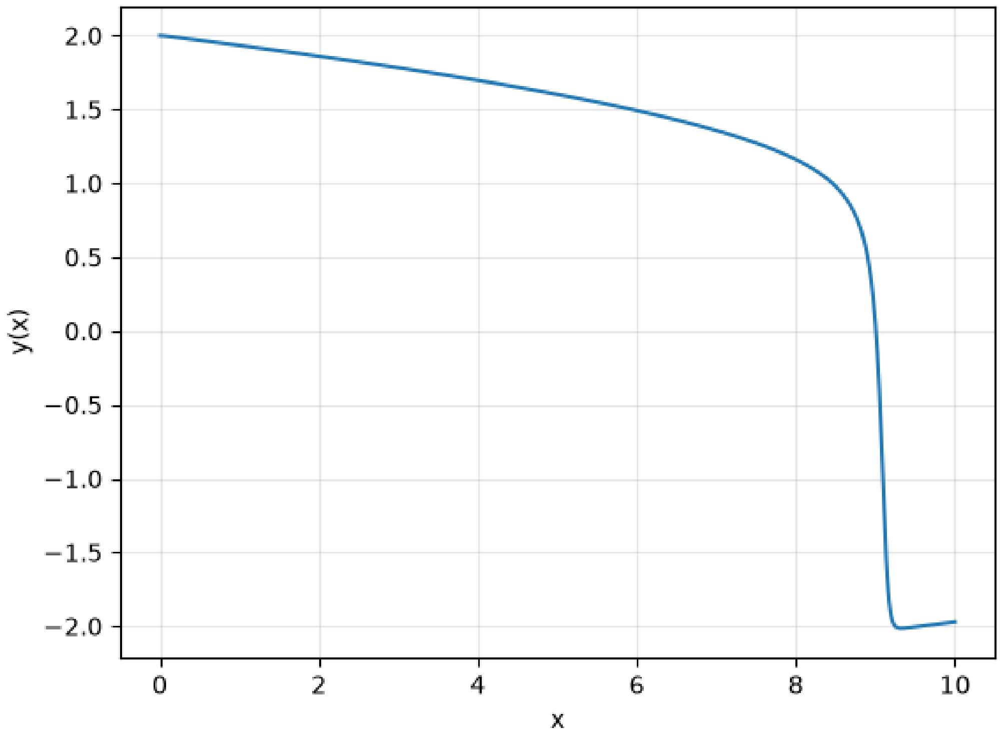

Figure 2: solution plot: $\mu = 10$.

$\mu = 100$

```
Piecewise power-series solution (328 segments of degree 20 covering [0, 10])

INFO: verification by substitution - VERIFIED: satisfies the equation
      to 2.34e-08 on [0, 10]. ... Interface continuity across 327
      node(s): max one-sided jump in (y..y1) = 3.0e-12 - measured,
      not assumed. Also checked: initial data to 0.00e+00.
solve time: 0.474 s
```

Ten times stiffer. Seven times more segments, and the accuracy does not degrade — the defect actually falls, and the 327 interface joins agree to three parts in 10^{12}.

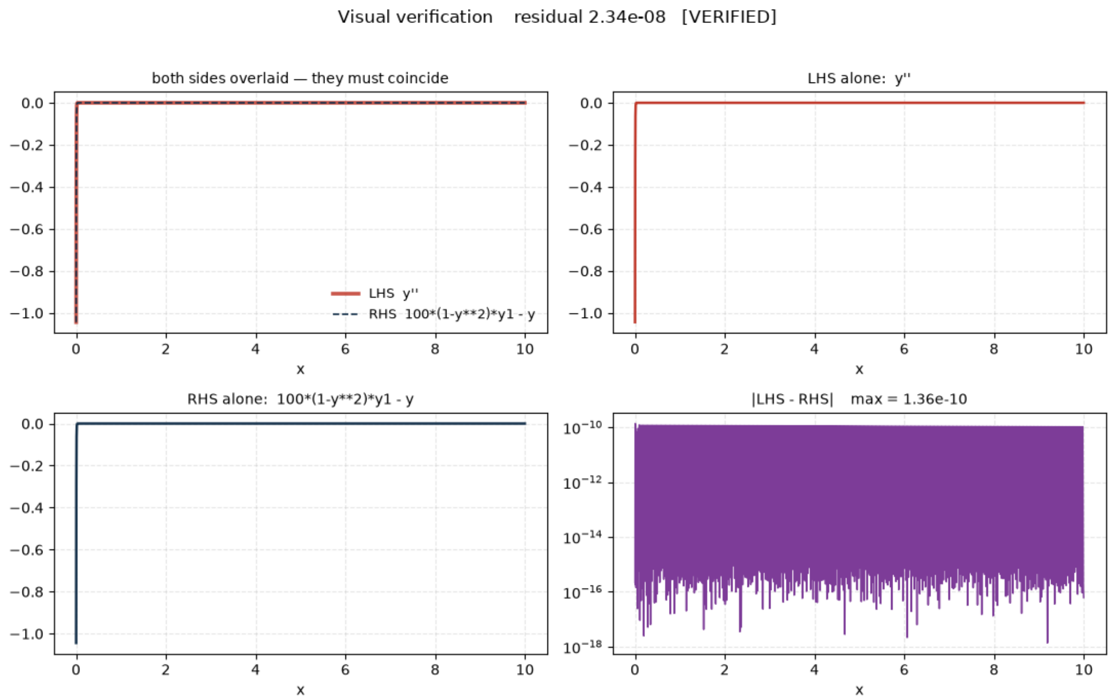

Figure 3: verification: $\mu = 100$.

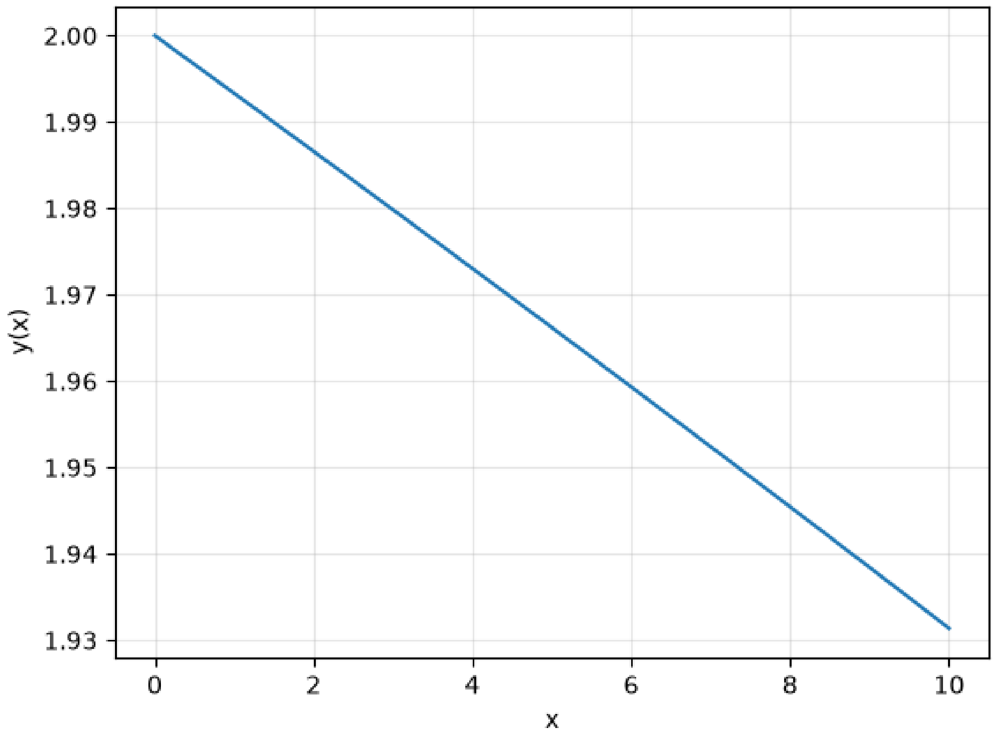

Figure 4: solution plot: $\mu = 100$.

$\mu = 1000$ — the part worth reading

Here the same call does something different, and explains itself before doing it:

```
INFO: the analytic march projects to ~3569 segments on this interval
      (step size limited by stiffness, not by any singularity); sirpy
      hands the march to its stall machinery immediately, whose
      classifier and labelled stiff delegation carry the honest
      result below.
INFO: the series march was PROGRESSING (0.05 s to x ~ 0.200293 of 10),
      but the stiffness pre-check had capped this attempt at 0.05 s:
      sirpy is switching strategy. At the measured rate, reaching 10
      would take about 3 s of pure series marching.
```

```
INFO: the series method could not reach 10, so the problem was handed
      to SciPy (Radau, rtol=1e-10). Solution by
      scipy.integrate.solve_ivp -- credit to the SciPy project. The
      result is a NUMERICAL grid, not a power series.
INFO: the delegated answer was AUDITED by substituting it back into
      the equation: max residual 7.00e-11 (relative 7.00e-11)
      -- consistent.
solve time: 0.570 s
```

Read what that block actually does.

It diagnoses. The steps are shrinking because of *stiffness, not a singularity* — two situations that look identical from the step size alone and demand opposite responses.

It estimates the bill before running it. ~3569 segments, about 3 seconds. That is a forecast, made after 0.05 seconds of measurement, of work it has not done.

It hands over, and says who to. Not "solved" — *"handed to SciPy (Radau, rtol=1e-10) — credit to the SciPy project"*, with the result labelled a **numerical grid, not a power series**. You always know which product you received.

It audits the borrowed answer. SIRPY substitutes SciPy's solution back into the original equation itself: residual 7×10^{-11}, *consistent*. Delegation is not trust.

It tells you how to overrule it. `method='sirpy'` for the pure-series attempt, `numeric_method='BDF'` or `'LSODA'` for a different scheme.

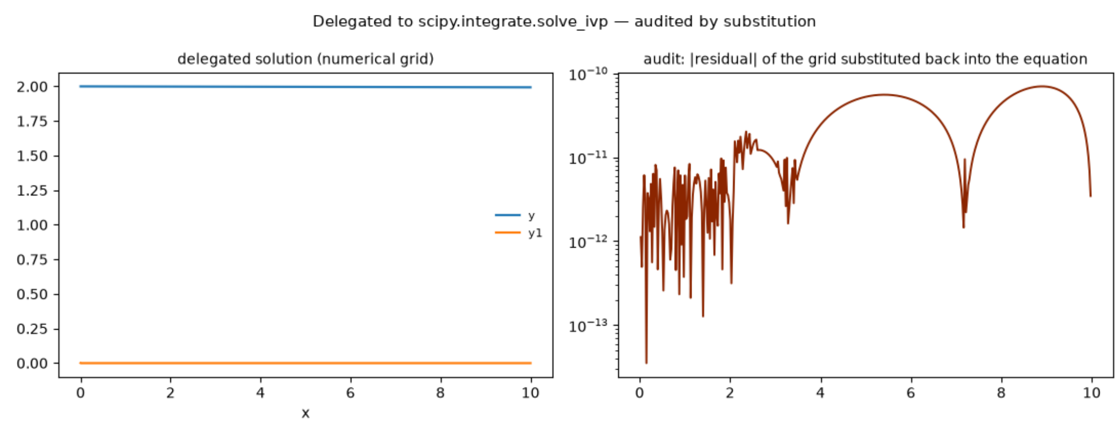

Figure 5: verification: $\mu = 1000$ - handed to SciPy.

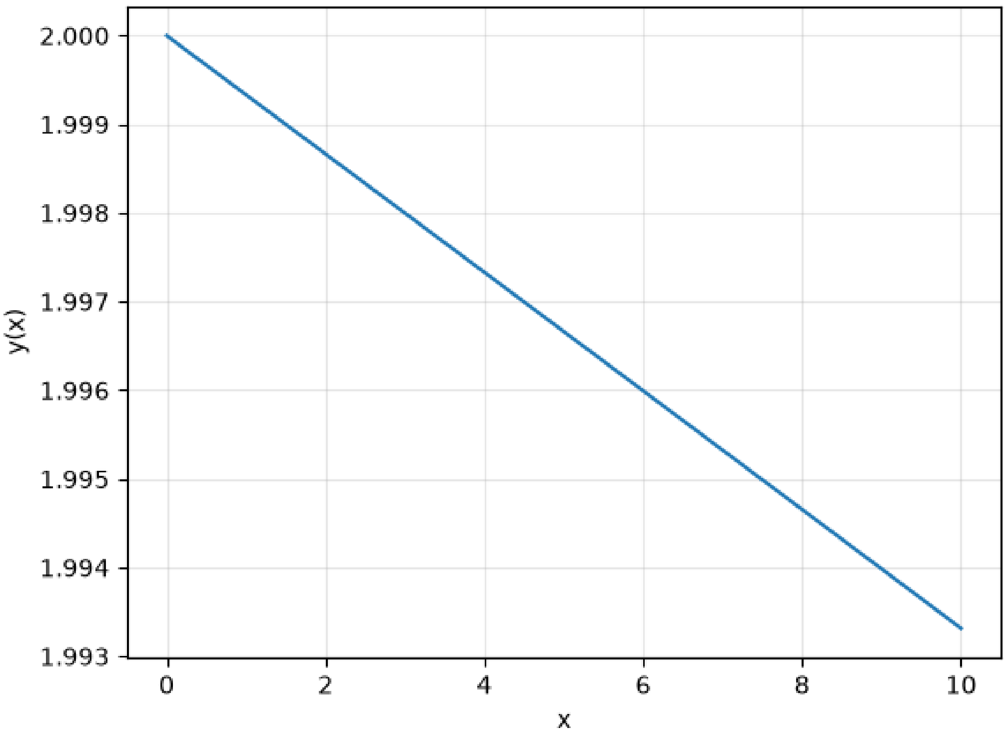

Figure 6: solution plot: $\mu = 1000$ - handed to SciPy.

Taking the override

We accepted the invitation:

```
solve_ivp(order=2, f_str="1000*(1 - y**2)*y1 - y",
          x0=0, x1=10, ic={0: 2, 1: 0}, method="sirpy")
```

```
Piecewise power-series solution (3388 segments of degree 20 covering [0, 10])

INFO: verification by substitution - VERIFIED: satisfies the equation
      to 2.09e-07 on [0, 10]. ... Interface continuity across 3387
      node(s): max one-sided jump in (y..y1) = 1.5e-12 - measured,
      not assumed. Also checked: initial data to 0.00e+00.
```

```
solve time: 8.489 s
```

3388 analytic segments across the stiff regime, every one of the 3387 joins measured. Both answers are available; they are different products, and nothing is hidden about which is which.

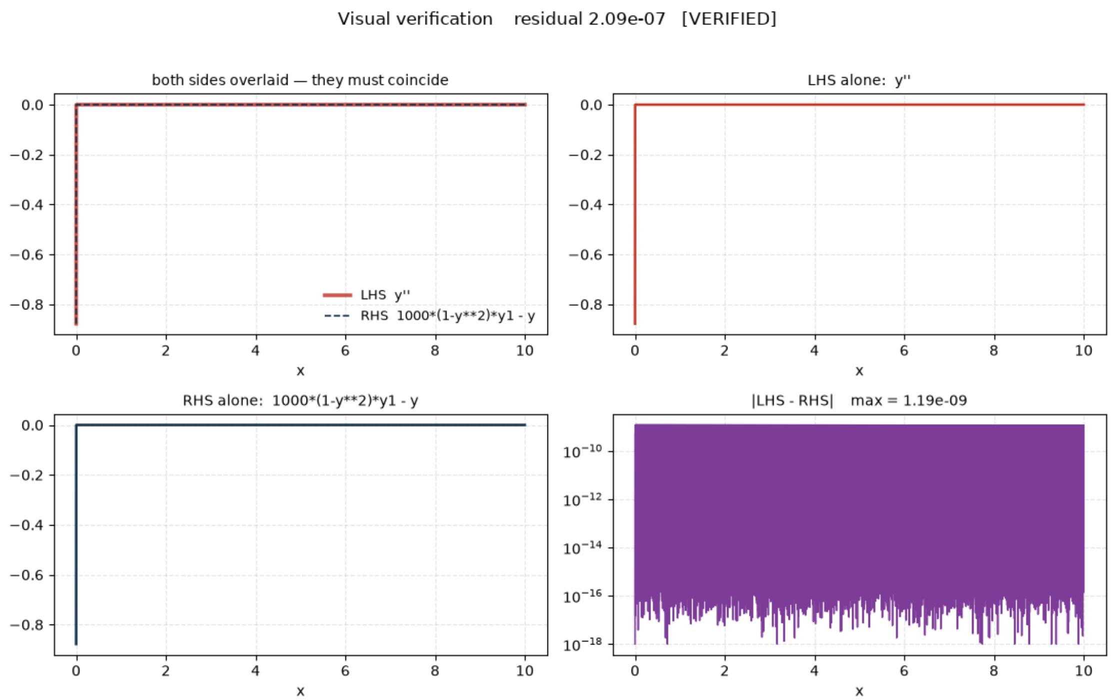

Figure 7: verification: $\mu = 1000$ - method="sirpy".

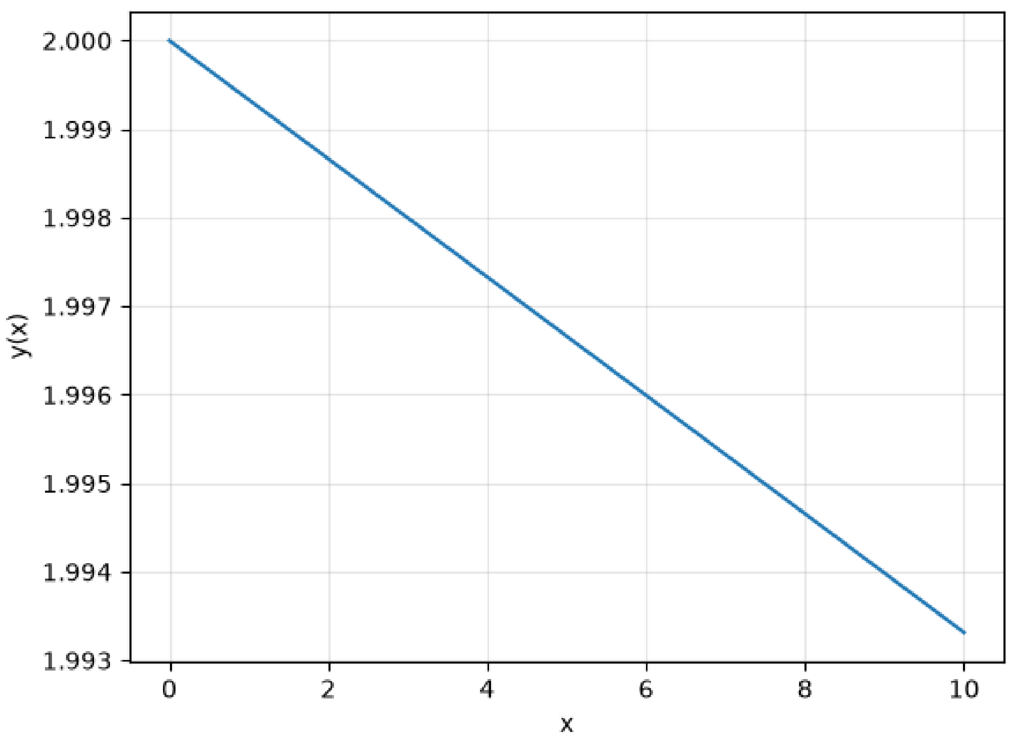

Figure 8: solution plot: $\mu = 1000$ - method="sirpy".

The whole study on one line each

μ	route	segments	solve time	equation defect	interface jump
10	series	46	0.130 s	6.17e-08 (rel 1.00e-08)	4.1e-10 over 45
100	series	328	0.474 s	2.34e-08	3.0e-12 over 327
1000	delegated, audited	numerical grid	0.570 s	7.00e-11 (audit)	—
1000	series (`method='sirpy'`)	3388	8.489 s	2.09e-07	1.5e-12 over 3387

Over a fixed interval the segment count tracks μ almost proportionally — 46, 328, 3388 for μ = 10, 100, 1000. That is the cost of stiffness, made visible rather than merely suffered. And the accuracy holds flat across a hundredfold change in stiffness: every defect stays between 10^{-8} and 10^{-7}.

The forecast, checked

SIRPY predicted ~**3569 segments** and **about 3 seconds**. The forced run took **3388 segments** and **8.489 seconds**.

The segment count was right to 5 %, from 0.05 seconds of measurement. The time estimate was optimistic by roughly a factor of three — because the projection extrapolates an early marching rate, and later segments are not free.

We report both because the forecast is the kind of claim that should be checked, and because a tool that tells you its estimate lets you check it. One that says nothing cannot be wrong out loud.

What this entry does and does not show

It shows cost, delegation behaviour, and verification across a hundredfold change in stiffness, on one fixed interval.

It does not show the full limit cycle at every μ. At $\mu = 1000$ the relaxation period is far longer than the interval [0, 10], so what is computed there — correctly, and verified — is the early motion. Showing complete oscillations at large μ is a different study, and we would rather say so than let a reader assume otherwise.

What this problem shows

Entered as written. One line, one changed number, across a hundredfold change in stiffness. No Jacobian, no tolerance, no stiffness flag.

Solved analytically where analytic is affordable. 46 and 328 exact Taylor pieces in fractions of a second; 3388 when asked for them explicitly.

Verified, not asserted. Equation defect by exact coefficient differentiation; relative residual reported when the equation's own scale is large; interface continuity measured at every join; initial data checked exactly.

Honest about cost before paying it. A stiffness diagnosis, a segment projection, and a time estimate — produced after a twentieth of a second, and checkable afterwards.

Honest about authorship. When SciPy solved it, SIRPY said so, credited the project, labelled the output a numerical grid rather than a power series — and then audited it anyway.

And it hands the decision back. Every delegation comes with the exact argument that overrules it.

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Contac Us

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SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from the solver's output; nothing is reproduced from memory.